

Using DDL Statements to Create and Manage Tables

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Naming Rules

- **Table names and column names:**
 - **Must begin with a letter**
 - **Must be 1–30 characters long**
 - **Must contain only A–Z, a–z, 0–9, _, \$, and #**
 - **Must not duplicate the name of another object owned by the same user**
 - **Must not be an Oracle server reserved word**

CREATE TABLE Statement

```
CREATE TABLE [schema.] table  
    (column datatype [DEFAULT expr] [, ...]);
```

- You specify:
 - Table name
 - Column name, column data type, and column size

```
CREATE TABLE dept  
    (deptno      NUMBER(2) ,  
     dname       VARCHAR2(14) ,  
     loc         VARCHAR2(13) ,  
     create_date DATE DEFAULT SYSDATE);
```

Including Constraints

- Constraints enforce rules at the table level.
- Constraints prevent the deletion of a table if there are dependencies.
- The following constraint types are valid:
 - NOT NULL
 - UNIQUE
 - PRIMARY KEY
 - FOREIGN KEY
 - CHECK



Defining Constraints

- Column-level constraint:

```
CREATE TABLE employees(  
    employee_id  NUMBER(6)  
    CONSTRAINT emp_emp_id_pk PRIMARY KEY,  
    first_name   VARCHAR2(20),  
    ...);
```

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- Table-level constraint:

```
CREATE TABLE employees(  
    employee_id  NUMBER(6),  
    first_name   VARCHAR2(20),  
    ...  
    job_id       VARCHAR2(10) NOT NULL,  
    CONSTRAINT emp_emp_id_pk  
        PRIMARY KEY (EMPLOYEE_ID));
```

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UNIQUE & CHECK Constraints

- Defined at either the table level or the column level:

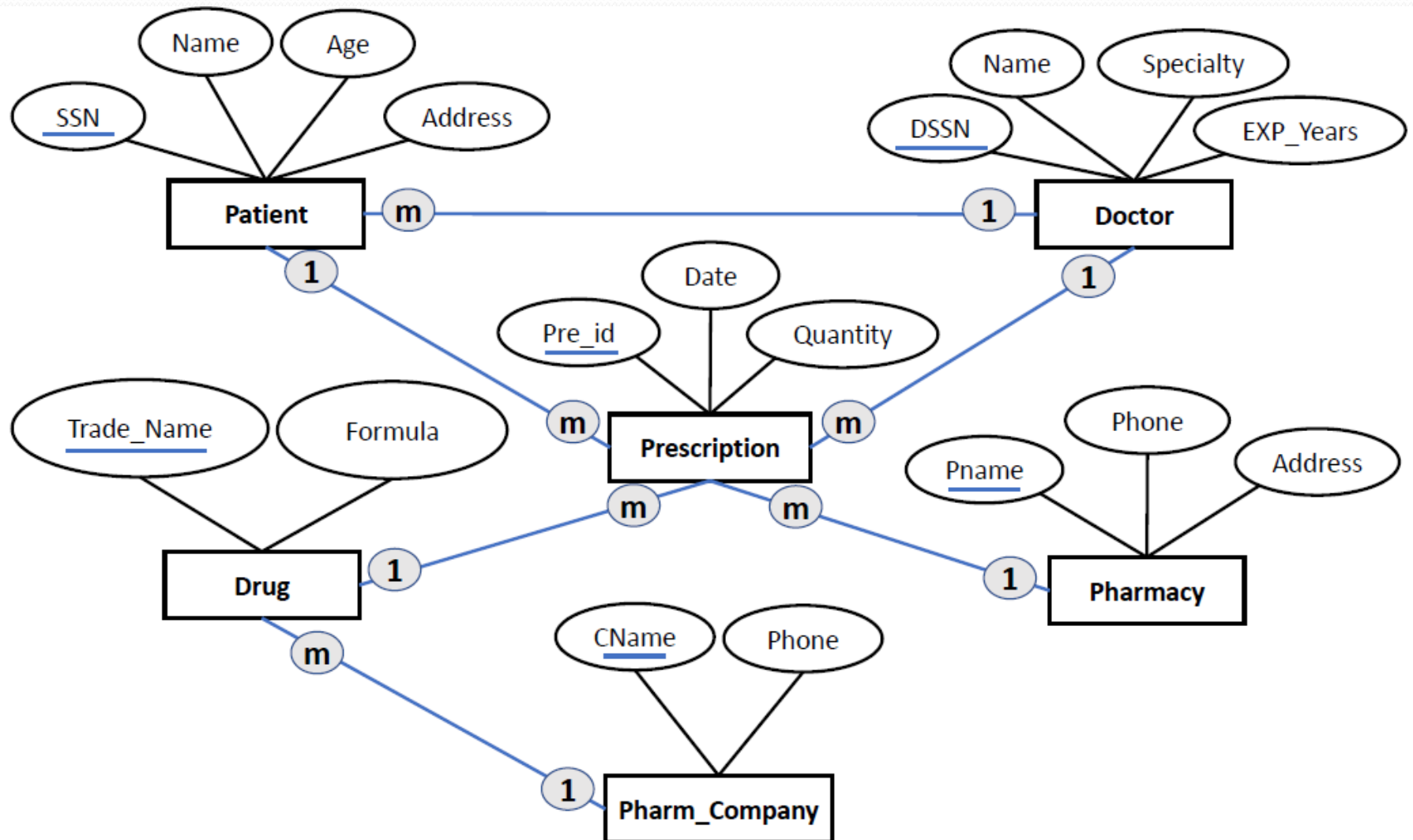
```
CREATE TABLE employees (  
    employee_id      NUMBER(6),  
    last_name        VARCHAR2(25) NOT NULL,  
    email            VARCHAR2(25),  
    salary            NUMBER(8,2),  
    commission_pct   NUMBER(2,2),  
    hire_date        DATE NOT NULL,  
  
    ...  
  
    CONSTRAINT emp_email_uk UNIQUE(email),  
    CONSTRAINT emp_salary_min CHECK (salary > 0)  
);
```

FOREIGN KEY Constraint

- Defined at either the table level or the column level:

```
CREATE TABLE employees (  
    employee_id      NUMBER(6) ,  
    last_name        VARCHAR2(25) NOT NULL ,  
    email            VARCHAR2(25) ,  
    salary            NUMBER(8,2) ,  
    commission_pct   NUMBER(2,2) ,  
    hire_date        DATE NOT NULL ,  
    ...  
    department_id    NUMBER(4) ,  
    CONSTRAINT emp_dept_fk FOREIGN KEY (department_id)  
        REFERENCES departments(department_id) ;
```

PharmacyERD



Task1

1. Write SQL statements to create all the required tables based on the provided Pharmacy ERD .
2. Insert to each table with 50 representative records using appropriate and realistic sample data.

```
CONNECT system/installedpassword;
```

```
-- for each section create a new user to build the same schema
```

```
CREATE USER Pharmcy IDENTIFIED BY pharmacy;  
GRANT DBA TO Pharmcy ;
```

```
CREATE TABLE doctor (  
  dssn    NUMBER PRIMARY KEY,  
  name    VARCHAR2(100) NOT NULL,  
  specialty VARCHAR2(100),  
  exp_years NUMBER  
);
```

```
CREATE TABLE patient (  
  ssn     NUMBER PRIMARY KEY,  
  name    VARCHAR2(100) NOT NULL,  
  age     NUMBER,  
  address VARCHAR2(200),  
  dssn    NUMBER,  
  CONSTRAINT fk_doctor_patient FOREIGN KEY ( dssn )  
    REFERENCES doctor ( dssn )  
);
```

```
CREATE TABLE pharmacy (  
  pname   VARCHAR2(100) PRIMARY KEY,  
  phone   VARCHAR2(20),  
  address VARCHAR2(200)  
);
```

```
CREATE TABLE Pharm_Company (  
  cname VARCHAR2(100) PRIMARY KEY,  
  phone VARCHAR2(20)  
);
```

```
CREATE TABLE Drug (  
    trade_name VARCHAR2(100),  
    formula VARCHAR2(100),  
    cname VARCHAR2(100),  
    PRIMARY KEY (trade_name, cname),  
    CONSTRAINT fk_drug_company FOREIGN KEY (cname)  
        REFERENCES Pharm_Company(cname)  
);
```

```
CREATE TABLE Prescription (  
    pre_id NUMBER PRIMARY KEY,  
    pre_date DATE,  
    quantity NUMBER,  
    patient_ssn NUMBER,  
    doctor_dssn NUMBER,  
    pharmacy_name VARCHAR2(100),  
    drug_trade_name VARCHAR2(100),  
    drug_cname VARCHAR2(100),  
    CONSTRAINT fk_presc_patient FOREIGN KEY (patient_ssn)  
        REFERENCES Patient(ssn),  
    CONSTRAINT fk_presc_doctor FOREIGN KEY (doctor_dssn)  
        REFERENCES Doctor(dssn),  
    CONSTRAINT fk_presc_pharmacy FOREIGN KEY (pharmacy_name)  
        REFERENCES Pharmacy(pname),  
    CONSTRAINT fk_presc_drug FOREIGN KEY (drug_trade_name, drug_cname)  
        REFERENCES Drug(trade_name, cname)  
);
```

Illustrate the ALTER statements

(Only Illustrate not Apply Code)

- If I need to add a new column to Doctor table

```
ALTER TABLE Doctor ADD (email VARCHAR2(100));
```

- If I need to modify column data type in Patient table

```
ALTER TABLE Patient MODIFY (address VARCHAR2(300));
```

- If I need to rename a column in Pharmacy table

```
ALTER TABLE Pharmacy RENAME COLUMN phone TO phone_number;
```

- If I need to add a new constraint (check constraint)

```
ALTER TABLE Patient ADD CONSTRAINT chk_age CHECK (age >= 0 AND age <= 120);
```

- If I need to add a foreign key from Drug to Pharm_Company (if not already exists)

```
ALTER TABLE Drug ADD CONSTRAINT fk_drug_company_new  
FOREIGN KEY (cname) REFERENCES Pharm_Company(cname);
```

- If I need to add a unique constraint

```
ALTER TABLE Doctor ADD CONSTRAINT uq_doctor_name UNIQUE (name);
```

Illustrate the DROP statements

(Only Illustrate not Apply Code)

- **if I need to drop a specific constraint**

```
ALTER TABLE Doctor DROP CONSTRAINT uq_doctor_name;
```

- **if I need to drop a column**

```
ALTER TABLE Doctor DROP COLUMN email;
```

- **if I need to drop tables**

```
DROP TABLE Prescription CASCADE CONSTRAINTS;  
DROP TABLE Patient CASCADE CONSTRAINTS;
```

- **if I need to drop the user and all objects**

```
CONNECT system/installedpassword;  
DROP USER Section1 CASCADE;
```

Insert 2 records in each table

- INSERT INTO Doctor VALUES (1001, 'Dr. Ahmed Ali', 'Cardiology', 25);
- INSERT INTO Doctor VALUES (1002, 'Dr. Mohamed Hassan', 'Dermatology', 8);
- INSERT INTO Pharm_Company VALUES ('Pharco (1)', '0226374122');
- INSERT INTO Pharm_Company VALUES ('Eva Pharma (2)', '0227707870');
- INSERT INTO Drug VALUES ('Panadol 500mg', 'Paracetamol 500mg', 'Pharco (1)');
- INSERT INTO Drug VALUES ('Augmentin 500mg', 'Omeprazole 20mg', 'Eva Pharma (2)');
- INSERT INTO Pharmacy VALUES ('El-Salam Pharmacy - Branch 1', '01186125617', 'Cairo, Egypt');
- INSERT INTO Pharmacy VALUES ('El-Salam Pharmacy - Branch 2', '01238538251', 'Alexandria, Egypt');
- INSERT INTO Patient VALUES (2001, 'Mohamed Ali', 27, 'Cairo, Egypt', 1001);
- INSERT INTO Patient VALUES (2002, 'Fatma Hassan', 25, 'Alexandria, Egypt', 1002);
- INSERT INTO Prescription VALUES (3001, TO_DATE('2025-09-21','YYYY-MM-DD'), 4, 2001, 1001, 'El-Salam Pharmacy - Branch 1', 'Panadol 500mg', 'Pharco (1));